Marco Silva Portuguese | Helsinki, Finland

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Work Experience

Senior Cybersecurity Software Developer — Ericsson 08/2025 – Present | Remote · Helsinki, Finland

- Platform Architecture: Development on ESM (Ericsson Security Manager), managing complex security posture, compliance, and telemetry for large-scale telecommunications networks.
- Microservices on Kubernetes: Architecting and maintaining Django microservices containerized and orchestrated on Kubernetes (`k8s` / `k3s`).
- High-Performance Reporting: Integrated automated reporting pipeline with Typst for high-quality, high-throughput PDF backend document generation.
- Framework Uplifts & Modernization: Leading major framework uplifts, dependency modernization, and structural refactoring across platform services.
- Dynamic Configuration: Implementing and managing Kubernetes `ConfigMaps` and secret injection patterns for runtime service configuration.
- Automated Testing: Designing end-to-end integration and system test suites using Robot Framework.
- AI Integration: Architectural planning and feature prototyping for AI-assisted capabilities within the platform.

Fullstack Developer — Black Duck 07/2019 – 08/2025 | Belfast, Northern Ireland, UK

- Internal Vulnerability Platform: Fullstack engineering of high-throughput internal platform used by security researchers and analysts to discover, triage, and score open-source software vulnerabilities.
- Unified Taxonomy System: Designed and implemented a unified taxonomy system categorizing security vulnerabilities, CWE/CAPEC patterns, and exploit vectors across diverse ecosystems.
- SCA Analysis in Rust: Developed a dedicated Software Composition Analysis (SCA) checker tool written in Rust for performance-critical manifest and binary parsing.
- Standards Integration: Embedded industry cybersecurity standards (OWASP, CWE, CAPEC, CVSS) into daily tooling and automated workflows.
- Testing Leadership: Introduced and built an extensive automated test suite from scratch using pytest and factory patterns, elevating code quality and reliability.
- Team Mentoring: Mentored team members on Python architecture, clean code practices, asynchronous programming, and REST API design.

Fullstack Developer & SCRUM Master — Hubel Positive Lightning 10/2015 – 07/2019 | Faro, Algarve, Portugal

- Leadership & Agile: Served as technical project lead and certified/practicing Agile SCRUM Master across cross-functional engineering teams.
- Real-Time IoT Backend: Engineered robust Python/Django backends utilizing Celery for distributed asynchronous workloads and Django Channels (WebSockets) for real-time telemetry streaming from industrial machinery.
- Industrial Systems: Deployed mission-critical monitoring and control platforms for industrial factories, commercial automated farms, and municipal water treatment facilities.
- LoRa & Hardware Integration: Managed long-range, low-power LoRa wireless networks and integrated remote telemetry sensors and PLCs.
- Fullstack Web Apps: Built interactive frontends in Angular and React, backed by structured REST APIs using Django REST Framework (DRF).
- Infrastructure & DevOps: Managed on-premise and cloud Linux server deployments, introducing containerization with Docker and Docker Compose.
- Mentorship: Guided university interns and junior engineers through practical software engineering, code reviews, and Git collaboration.

Education

Postgraduate Diploma in Cybersecurity — University of the Algarve (Universidade do Algarve) 09/2024 – 06/2025 | Faro, Portugal

- Cryptographic algorithms, public key infrastructure (PKI), TLS/SSL protocols.
- Secure software development lifecycle (SSDLC), threat modeling, and defense-in-depth principles.
- Penetration testing methodology, network security assessments, vulnerability scanning.
- Incident detection, SIEM log analysis, containment, malware triage, and digital forensics.
- Security standards and frameworks (ISO 27001, NIST CSF, NIS2, GDPR).
- Risk management matrices, compliance auditing, business continuity planning.

Master in Electronics and Telecommunications Engineering (Coursework completed) — University of the Algarve (Universidade do Algarve) 09/2009 – 09/2015 | Faro, Portugal

- Computer Science & Software: Algorithms, data structures, object-oriented programming, operating systems, embedded systems C/C++.
- Signals & Control Systems: Discrete-time signal processing (DSP), linear dynamical systems, digital filtering, control theory.
- Telecommunications & Networks: RF and antenna design, modulation schemes, TCP/IP networking, wireless transmission systems, cellular network architectures.
- Machine Learning & AI: Neural networks, pattern recognition, regression models, supervised/ unsupervised learning.

Core Technical Skills

Django, DRF, Celery, Python, Angular, React, JavaScript, Rust, Kubernetes, Helm, Docker, Robot Framework, Neovim, Git

Competitive Achievements & CTFs

- HTB Business CTF 2025: Global rank #19 | HTB Business CTF 2024: #14 | HTB Business CTF 2023: #12
- Cyber Apocalypse CTF: 2025 (#86) | 2024 (#38) | 2023 (#15)
- Synopsys CSAM 2022: #2
- Advent of Code: 2024 (50 stars) | 2023 (42 stars) | 2022 (40 stars)

Languages

- Portuguese: Native
- English: C2 Proficient
- Spanish: B1 Intermediate